

Virtual Memory and Paging in Operating System

1. Virtual Memory:

Virtual memory allows execution of programs larger than RAM by using disk as extended memory.

How it works:

- Only required parts loaded into RAM
- Remaining stored in disk
- Data swapped when needed

Idea:

Gives illusion of large memory

2. Paging:

- Memory divided into pages (logical) and frames (physical)
- Page table maps pages to frames

Address Translation:

Logical Address = Page Number + Offset

Steps:

1. CPU generates address
2. Page table checked
3. Frame found
4. Physical address created

Advantages:

- Runs large programs
- Efficient memory usage
- Multitasking
- No external fragmentation

Disadvantages:

- Slow performance
- Page table overhead
- Internal fragmentation